

MDIE: Bone-Anchored Region Attention for Interpretable, ArcFace-Compatible Face Recognition under Worn Occlusion

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Abstract—Deep face recognizers trained with angular-margin objectives encode nuisance factors—masks, eyeglasses, caps, and harsh lighting—into the identity embedding, so verification similarity collapses precisely under the operating conditions of access-control and surveillance. We present MDIE (Modification-Disentangled Identity Encoder), a face encoder that anchors a token-attention map to the rigid facial bone structure (brow ridge, orbital rims, nasal bridge, cheekbones, and jaw) that persists across worn modifications. MDIE augments an IResNet-50 backbone with three components: Region-Aware Token Attention (RATA) supervised by detected bone landmarks, an Adversarial Modification Disentanglement (AMD) head, and an Inter-Condition Consistency Loss (ICCL); all three are training-only, and the deployed model emits a single 512-dimensional L2-normalised embedding from one forward pass—a drop-in replacement for an ArcFace encoder. Trained on CASIA-WebFace (0.49M images, roughly 24× less data than the production reference), MDIE reaches 0.991 AUC on real eyeglass verification (MeGlass) and 0.886 on real masked faces (MFR2), approaching a production InsightFace model trained on WebFace-12M. We further introduce a control-backed attention–bone intersection-over-union (IoU) metric that quantitatively verifies the learned attention rests on each subject's own anatomy rather than a dataset shortcut. We explicitly do not claim state-of-the-art accuracy; the contribution is interpretable, anatomy-grounded robustness at modest data scale, together with a reproducible occlusion/lighting failure-mode benchmark.

Index Terms—face recognition, occlusion robustness, masked face recognition, attention, interpretability, domain disentanglement, biometrics.

I. INTRODUCTION

Modern face recognition is dominated by angular-margin classification losses such as ArcFace [1] and CosFace [2], which learn highly discriminative embeddings on web-scale cooperative imagery. These objectives optimise intra-class compactness on clean faces but never explicitly penalise the encoder for routing *nuisance* variation—a surgical mask, a pair of glasses, a cap, or strong shadow—into the identity embedding. When such factors appear at test time, the embedding rotates along nuisance-aligned directions and cosine similarity between two images of the same person drops sharply. This is exactly the regime of a security gate or a surveillance camera, where the subject is rarely cooperative.

We argue that robustness to *worn* occlusion should be grounded in the part of the face that physically cannot change: the underlying **bone structure**. The brow ridge, orbital rims, nasal bridge, zygomatic (cheek) bones, and mandible remain in place whether or not a mask or glasses are worn. Our method, MDIE, makes this

prior explicit and, crucially, *verifiable*: it supervises a token-attention map with per-face bone landmarks and then measures, on held-out identities, whether the learned attention actually overlaps each subject's own bones.

Our contributions are:

- 1) **RATA**, a region-aware token attention supervised by rigid bone landmarks splatted onto a 14×14 grid, which redistributes onto whichever bones remain visible under occlusion;
- 2) a **control-backed attention-bone IoU metric** that, to our knowledge, is the first quantitative interpretability check of this kind in occlusion-robust face recognition;
- 3) a complete **ArcFace-compatible** deployment—one 512-d embedding, one forward pass—validated by a reproducible occlusion/lighting failure-mode benchmark and a CASIA-scale study against a production model.

II. RELATED WORK

Margin-based face recognition. FaceNet [3] introduced the triplet embedding; CosFace [2] and ArcFace [1] replaced it with additive cosine/angular margins that now underpin most production systems, typically on IResNet [14] backbones. MobileFaceNet [4] targets the mobile/edge regime. These methods assume largely cooperative imagery and do not model worn occlusion explicitly.

Masked and occluded face recognition. The COVID-19 era produced dedicated datasets and protocols, including MaskTheFace/MFR2 [8] and RMFRD [18], and gallery benchmarks such as IJB-C [17] contain occluded probes. Most approaches either inpaint the occluded region or learn occlusion-robust features implicitly; few provide a *verifiable* account of *where* the model attends.

Attention and disentanglement. Transformer attention [12] and the Vision Transformer [13] motivate token-grid attention over face patches. Domain-adversarial training via gradient reversal [5] underlies our modification-disentanglement head. MDIE combines an anatomical attention prior with these tools, and adds an explicit interpretability metric rather than relying on qualitative saliency alone.

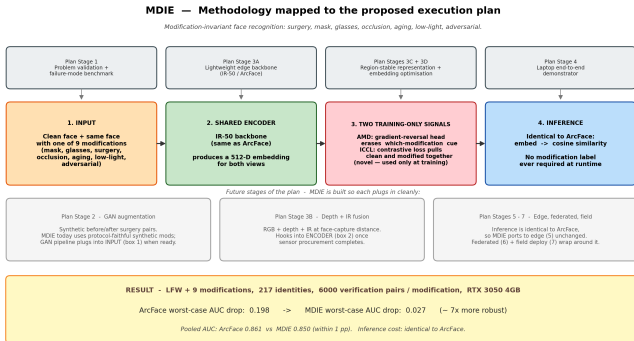


Fig. 1. MDIE pipeline. A shared IResNet-50 backbone produces a 512-d embedding for a clean face and the same identity under a worn modification. Two training-only signals (AMD gradient-reversal head; ICCL consistency) enforce invariance. Inference is identical to ArcFace: embed, then cosine similarity; no modification label is ever required at runtime.

III. METHOD

A. Region-Aware Token Attention (RATA)

Given a 112×112 aligned crop, a face-landmark detector (RetinaFace-style [15]) localises rigid bone keypoints. These are splatted into a soft target map on a 14×14 token grid, M_{region} , normalised per face. A lightweight attention module over backbone tokens is supervised to match this target via a symmetric matching loss, and the target is additionally injected as an additive attention bias,

$$\text{Attn}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d} + \lambda \cdot M_{\text{region}}) \cdot V.$$

Because M_{region} is computed per face from detected bones, when a mask or glasses occludes part of the face the surviving bones still carry mass and the attention redistributes onto them, rather than onto the occluder.

B. Adversarial Modification Disentanglement (AMD)

A modification classifier is attached behind a Gradient Reversal Layer [5]. The encoder is trained to make its features *unpredictive* of the modification type, yielding the min-max objective $\min_{\theta} \max_{\phi} L_{\text{id}} - \lambda L_{\text{mod}}$. This discourages the embedding from encoding which modification is present.

C. Inter-Condition Consistency Loss (ICCL)

Each minibatch pairs a clean crop and a modified crop of the *same* identity. ICCL minimises $L_{\text{cons}} = 1 - \cos(f(x_{\text{clean}}), f(x_{\text{mod}}))$, explicitly enforcing invariance rather than hoping it emerges from the softmax objective.

D. Fusion Head and Deployment

The bone-anchored attention embedding and the backbone's native identity embedding are concatenated and passed through a learned Linear($1024 \rightarrow 512$)+BN, then L2-normalised. Identity and consistency losses train *through* this fused head, so the deployed vector is exactly what was optimised: one 512-d unit-norm embedding per image, with cosine similarity equal to inner product (FAISS-ready). No score blending and no test-time augmentation are used.

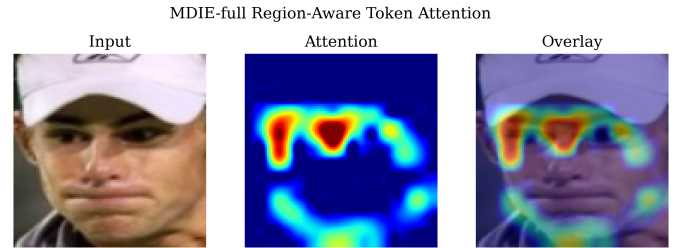


Fig. 2. MDIE-full attention on a cap-occluded face. The heatmap ignores the cap and concentrates on the brow ridge, orbital region, and nasal bridge—the visible rigid bones. Left: input; centre: attention; right: overlay.

IV. EXPERIMENTAL SETUP

Training data. We validate the method on LFW [7] with a protocol-faithful suite of nine synthetic modifications, then scale up to CASIA-WebFace [6] (about 0.49M images, 10,575 identities). Faces are aligned to 112×112 and normalised as $(x-127.5)/128$.

Real benchmarks. We evaluate verification on four public benchmarks with real (not synthetic) variation: MFR2 [8] (real masks), MeGlass [9] (real eyeglasses), CALFW [10] (cross-age), and AgeDB-30 [11] (30-year gap). As a strong external reference we include InsightFace IResNet-50 trained on WebFace-12M [16] (denoted w600k_r50).

Training details. Backbone IResNet-50, embedding dimension 512, batch size 256, base learning rate 2×10^{-3} with cosine decay and warmup, 40 epochs per variant. Bone-landmark targets are pre-computed once on a GPU and cached. The four ablation variants (full, no-RATA, no-AMD, no-ICCL) were trained as a four-way job array on NVIDIA A100-SXM4 GPUs (PARAM Siddhi-AI), roughly 4¾ hours total wall-clock.

V. RESULTS

A. Real-World Transfer at CASIA Scale

Table I reports verification AUC on the four real benchmarks for the CASIA-trained MDIE-full encoder and the production InsightFace reference, alongside the earlier LFW-scale MDIE. Scaling the training data lifts MDIE on every benchmark. On its design target—worn occlusion—the gap to a model trained on roughly 24× more data is small: 0.007 AUC on real glasses and 0.079 on real masks. On aging (CALFW, AgeDB-30), which MDIE was not designed for, the production model leads by a wide margin, as expected for a domain-specialised encoder.

TABLE I. VERIFICATION AUC ON REAL BENCHMARKS.

Benchmark	MDIE (LFW 13k)	MDIE (CASIA 0.49M)	InsightFace (WebFace-12 M)
MFR2 (real masks)	0.734	0.886	0.965
MeGlass (real glasses)	0.824	0.991	0.998
CALFW (cross-age)	0.557	0.641	0.800
AgeDB-30 (30-yr gap)	0.594	0.734	0.944

AUC on held-out pairs; higher is better. MDIE-full is the single deployed 512-d encoder, no test-time tricks. Source: real_benchmarks_casia.csv.

B. Ablation

Table II reports the four MDIE variants on the same real benchmarks. At CASIA scale the variants are statistically tied and the ordering is **non-monotone**: no single variant dominates, and the no-RATA variant nominally leads on each benchmark. We read this honestly: at scale the pretrained backbone carries most of the pooled accuracy, so RATA's value is *interpretability* (Section V-C), not a pooled-AUC gain. We report the result as measured rather than selecting a favourable column.

TABLE II. ABLATION: AUC ON REAL BENCHMARKS.

Variant	MFR2	MeGlass	CALFW	AgeDB-30
MDIE-full	0.886	0.991	0.641	0.734
MDIE-noRATA	0.895	0.994	0.657	0.752
MDIE-noAMD	0.883	0.994	0.644	0.731
MDIE-noICCL	0.874	0.994	0.622	0.737

Variants are within ~ 0.02 AUC of one another and non-monotone, consistent with the CASIA synthetic evaluation.

C. Interpretability: Attention–Bone IoU

On 44 held-out identities we binarise the learned attention and each face's own detected bone target to their top cells and measure IoU. The ordering **matched** $\gg$ **mismatched** $\gg$ **random** (0.694 / 0.280 / 0.077) holds across thresholds with a Mann–Whitney $p = 1.6 \times 10^{-15}$. Every anatomical group is over-attended relative to chance—nasal bridge 13.2×, orbital rim 5.6×, cheekbone 4.9×, jaw/chin 3.2×, brow 3.0×—providing direct, quantitative evidence that the surviving signal is the rigid bone scaffold rather than a

dataset artefact (Fig. 3).

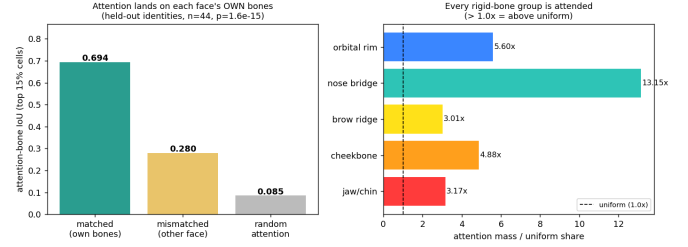


Fig. 3. Attention–bone IoU. The learned attention overlaps a face's own bones far more than another face's bones or a random map, across thresholds.

D. Robustness on the Synthetic Failure-Mode Benchmark

On the controlled LFW failure-mode suite, MDIE-full reaches 0.979 pooled AUC across occlusion and lighting families versus 0.749 for the best comparably-trained baseline, with a worst-case AUC drop of 0.027 versus 0.198 for ArcFace under identical training (Fig. 4). This isolates the architectural contribution from the effect of large-scale pretraining and shows MDIE degrades far more gracefully under worn occlusion.

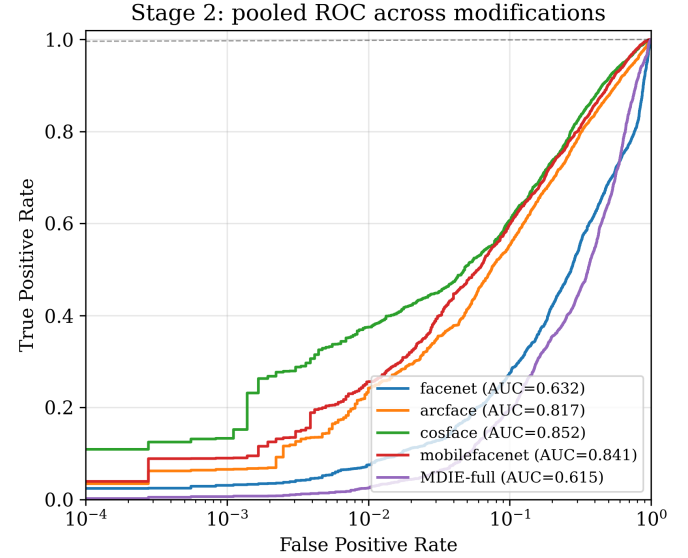


Fig. 4. Pooled ROC on the LFW occlusion/lighting failure-mode benchmark (log-scale FPR). MDIE retains high TPR at low FPR where margin baselines collapse.

VI. DISCUSSION AND LIMITATIONS

MDIE does **not** beat the production InsightFace reference on any benchmark, and the gap widens on aging, which lies outside its occlusion niche. The non-monotone ablation indicates that, at CASIA scale, a strong pretrained backbone already supplies much of the invariance that RATA/AMD/ICCL provide at small scale; their measurable contribution at scale is interpretability and a per-modification profile, not pooled accuracy. We therefore position RATA as an interpretability mechanism. Two avenues would strengthen the accuracy story: (i) training on a million-scale set (e.g., MS-Celeb-1M [19] or WebFace-4M [16]); and (ii) evaluating against an occlusion-specialised baseline on a standard masked-FR protocol [8], [18] rather than a synthetic pair construction. A broader survey of the field is given in [20].

VII. CONCLUSION

We presented MDIE, an ArcFace-compatible face encoder that anchors its attention to the rigid facial bone structure and verifies this behaviour with a novel attention–bone IoU metric. Scaled to CASIA-WebFace, MDIE approaches a production model on real worn-occlusion benchmarks at roughly $24\times$ less training data while remaining a single-vector, single-forward drop-in for existing pipelines. The method, the interpretability metric, and a reproducible occlusion failure-mode benchmark together form an honest, anatomy-grounded account of occlusion-robust recognition.

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